Limitations: Guiding Questions

Having carried out this assignment, please write two paragraphs about the inherent limitations of carrying out analytics over anonymously submitted data items. Did the analytic responses surprise you? How does this different from standards? For example, the average GRE quantitative reasoning score was 157 for 2023-2023 and was nearly 165 for grad school entries submitted (see sample output). Why do you think that is? What might cause this to occur? Please place your essay into a file called limitations.pdf

Answers:

There are many inherent limitations with carrying out analysts with anonymously submitted data items. The first limitation I noticed is that some of the meaningful quantitative scores like GRE were optional fields for submission. This meant that when I drew averages I was missing a large subset of potential data that could have better informed my conclusions. The other obvious but impactful limitation is that free text fields allow users to enter a large variety of data as their program or university name. While the LLM can help to mediate some of the parsing issues that such data presents, some plausibly useful data points will always remain meaningfully unusable in a massive data set like the one we are analyzing.

The analytic responses surprised me because the GRE average for my analysis was 216. I know that my data is skewed because my computer was struggling to run the LLM processor on all 49000 rows that I was able to extract from the website but regardless I checked the minimum and maximum and found that the range indicated that some people had submitted section scores as total GRE scores skewing results. I think the difference between the submitted GRE was higher because people are less inclined to post lower scores or they may be posting the highest among all the times that sat for the exam, also skewing results. While this analysis is beneficial, I do believe there are more reliable data sources to use like from the Department of Education for example to understand similar data about grad school applications.